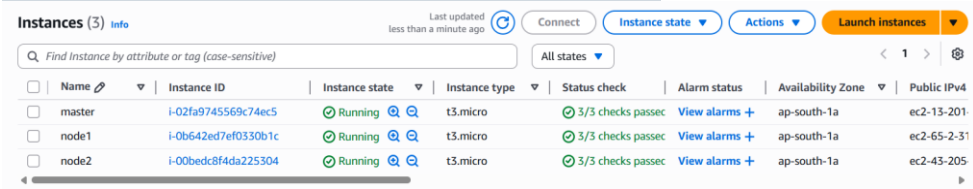
Project Name: Docker Swarm- Voting application

<table>
<tr><td>Objective</td><td>To deploy a scalable, multi-service voting application on a manager node, ensuring efficient orchestration, fault tolerance, and seamless monitoring through Docker visualizer</td></tr>
<tr><td>Problem Statement and Motivation</td><td>John, a DevOps engineer, is tasked with deploying a voting application through multiple microservices. By creating a Docker compose file and deploying it on a manager node in a distributed system, they ensure that each service is efficiently orchestrated and fault-tolerant.

To monitor the deployment, John integrates Docker visualizer as a microservice, providing real-time insights. This setup simplifies the deployment process, enhances scalability, and ensures the application runs smoothly in a production environment.</td></tr>
<tr><td>Create EC2 instances</td><td>
<table>
<tr><th>node</th><th>public-ip</th><th>private-ip</th></tr>
<tr><td>master</td><td>13.201.84.118</td><td>172.31.40.230</td></tr>
<tr><td>ndoe1</td><td>65.2.31.175</td><td>172.31.42.42</td></tr>
<tr><td>node2</td><td>43.205.127.233</td><td>172.31.43.203</td></tr>
</table>


</td></tr>
<tr><td>Install docker on all 3 instances</td><td>sudo su
apt update -y
apt install docker.io -y</td></tr>
<tr><td>Install docker swarm on master node</td><td># master: 172.31.40.230
docker swarm init --advertise-addr 172.31.40.230

#note down node join command</td></tr>
<tr><td>join node1 and node2 in swarm cluster</td><td>#node join command
docker swarm join --token SWMTKN-1-1s3jiuk6saa2ps6h5602zdz2gko6sfgusm2nx7n2d4jq9aftt6-0zjceti04fntt7j3dwft5pmzx 172.31.40.230:2377</td></tr>
<tr><td>overview of voting website</td><td>• Home page with 4 political party buttons
• Vote page that saves the vote in a free MySQL database
• Results page to see totals</td></tr>
</table>

	• Uses PHP + MySQL (works on free hosting like InfinityFree, 000webhost, or Hostinger free plans)
Structure:	voting-app/ ├── Dockerfile ├── index.php ├── vote.php ├── results.php └── db.php
Dockerfile	FROM php:8.2-apache RUN apt update -y RUN apt install git -y RUN git clone https://github.com/pradeepviswa/voting-app.git /var/www/html RUN ls /var/www/html RUN docker-php-ext-install mysqli #check mysqli RUN php -i | grep mysqli
Build Docker Image	docker build -t voting-image .
Create Overlay Network	docker network create \ --driver overlay \ voting-network
Check new network	docker network ls root@ip-172-31-40-230:/home/ubuntu# docker network ls NETWORK ID NAME DRIVER SCOPE bb9bb550470b bridge bridge local 9370616e072c docker_gwbridge bridge local 0d547bd63236 host host local 6zlua77e2ae8 ingress overlay swarm d9a8fede5bdd none null local k714jart0cs4 voting-network overlay swarm
Create MySQL Service	docker service create \ --name voting-mysql \ --network voting-network \ -e MYSQL_ROOT_PASSWORD=root \ -e MYSQL_DATABASE=voting_db \ mysql:8 root@ip-172-31-40-230:/home/ubuntu# docker service create \ --name voting-mysql \ --network voting-network \ -e MYSQL_ROOT_PASSWORD=root \ -e MYSQL_DATABASE=voting_db \ mysql:8 c3dyn923bzxahxt5b8xx0mexm overall progress: 1 out of 1 tasks 1/1: running [==========>] verify: Service c3dyn923bzxahxt5b8xx0mexm converged
Create Web Service	docker service create \ --name voting-web \ --network voting-network \ -p 8080:80 \ voting-image

```
root@ip-172-31-40-230:/home/ubuntu# docker service create \
> --name voting-web \
> --network voting-network \
> -p 8080:80 \
> voting-image
image voting-image:latest could not be accessed on a registry to record
its digest. Each node will access voting-image:latest independently,
possibly leading to different nodes running different
versions of the image.

oxql4oz6bow530h5uc55yb6mv
overall progress: 1 out of 1 tasks
1/1: running   [==================================================>]
verify: Service oxql4oz6bow530h5uc55yb6mv converged
```

Check service:

docker service ls

```
root@ip-172-31-40-230:/home/ubuntu# docker service ls
ID             NAME           MODE         REPLICAS   IMAGE                 PORTS
c3dyn923bzxa   voting-mysql   replicated   1/1        mysql:8
oxql4oz6bow5   voting-web     replicated   1/1        voting-image:latest   *:8080->80/tcp
```

Database Connection

My db.php uses the service name as hostname:
$host = "voting-mysql";
Swarm DNS automatically resolves the service.

Enter MySQL Service to Create Table

get container ID: docker ps
Connect mysql: docker exec -it <mysql_container_id> mysql -u root -p
#password is root
Create table:

```
USE voting_db;

CREATE TABLE votes (
id INT AUTO_INCREMENT PRIMARY KEY,
party VARCHAR(50)
);
```

```
root@ip-172-31-42-42:/home/ubuntu# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS         PORTS                    NAMES
a226f6e86b27   mysql:8   "docker-entrypoint.s…"   4 minutes ago   Up 4 minutes   3306/tcp, 33060/tcp      voting-mysql.1.qsjoesj1r04l74gpiwrs37hv6
root@ip-172-31-42-42:/home/ubuntu# docker exec -it a226f6e86b27 mysql -u root -p root
Enter password:
ERROR 1049 (42000): Unknown database 'root'
root@ip-172-31-42-42:/home/ubuntu# docker exec -it a226f6e86b27 mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 8.4.8 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE voting_db;
CREATE TABLE votes (
id INT AUTO_INCREMENT PRIMARY KEY,
party VARCHAR(50)
);
Database changed
mysql>
mysql> CREATE TABLE votes (
    -> id INT AUTO_INCREMENT PRIMARY KEY,
    -> party VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.04 sec)
```
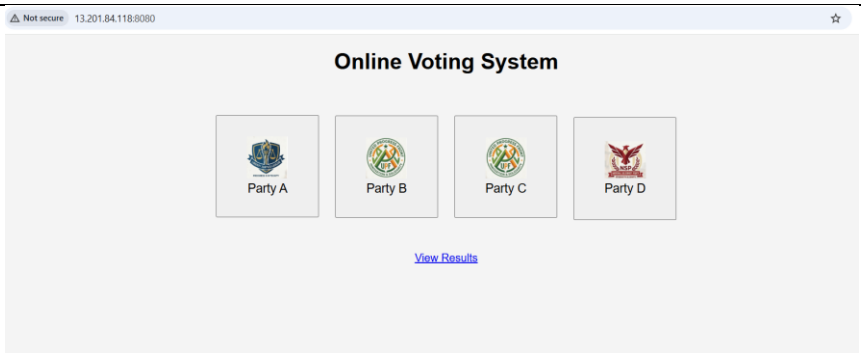

Scale the Web Service (Swarm advantage)

docker service scale voting-web=3

```
root@ip-172-31-40-230:/home/ubuntu# docker service scale voting-web=3
voting-web scaled to 3
overall progress: 3 out of 3 tasks
1/3: running   [==================================================>]
2/3: running   [==================================================>]
3/3: running   [==================================================>]
verify: Service voting-web converged
root@ip-172-31-40-230:/home/ubuntu# docker service ps voting-web
ID             NAME              IMAGE                NODE                DESIRED STATE   CURRENT STATE            ERROR                              PORT
S
qnu6uml02wjs   voting-web.1      voting-image:latest  ip-172-31-40-230    Running         Running 23 minutes ago
htuhsqdslyj5   voting-web.2      voting-image:latest  ip-172-31-40-230    Running         Running 36 seconds ago
vuptbom8w1ik    \_ voting-web.2  voting-image:latest  ip-172-31-43-203    Shutdown        Rejected 44 seconds ago   "No such image: voting-image:l…"
uorsq5q8k7t3    \_ voting-web.2  voting-image:latest  ip-172-31-43-203    Shutdown        Rejected 49 seconds ago   "No such image: voting-image:l…"
kxnuax1grba6    \_ voting-web.2  voting-image:latest  ip-172-31-43-203    Shutdown        Rejected 54 seconds ago   "No such image: voting-image:l…"
q1nd3dvlzgds    \_ voting-web.2  voting-image:latest  ip-172-31-42-42     Shutdown        Rejected 59 seconds ago   "No such image: voting-image:l…"
t7nci6y3y797   voting-web.3      voting-image:latest  ip-172-31-40-230    Running         Running 36 seconds ago
02ygty6j3v90    \_ voting-web.3  voting-image:latest  ip-172-31-42-42     Shutdown        Rejected 44 seconds ago   "No such image: voting-image:l…"
tsjvhbqzzeb2    \_ voting-web.3  voting-image:latest  ip-172-31-42-42     Shutdown        Rejected 49 seconds ago   "No such image: voting-image:l…"
vv9zzdvl54wz    \_ voting-web.3  voting-image:latest  ip-172-31-42-42     Shutdown        Rejected 54 seconds ago   "No such image: voting-image:l…"
4ldfj6yetxng    \_ voting-web.3  voting-image:latest  ip-172-31-43-203    Shutdown        Rejected 59 seconds ago   "No such image: voting-image:l…"
root@ip-172-31-40-230:/home/ubuntu#
```

Output:

Open Website And Click on different buttons multiple times	
When clicked on any button here is the output	
See Results page	

How voting app website is configured?

Git location	https://github.com/pradeepviswa/voting-app.git

db.php	<?php
	
	$host = "voting-mysql";
	$user = "root";
	$password = "root";
	$database = "voting_db";
	
	$conn = new mysqli($host,$user,$password,$database);
	
	if ($conn->connect_error) {
	 die("Connection failed: " . $conn->connect_error);
	}
	
	?>

index.html	<!DOCTYPE html> <html> <head> <title>Online Voting System</title> <link rel="stylesheet" href="style.css"> </head> <body> <h1>Online Voting System</h1> <form action="vote.php" method="POST"> <button name="party" value="Party A"> <img src="img/party1.png"> Party A </button> <button name="party" value="Party B"> <img src="img/party2.png"> Party B </button> <button name="party" value="Party C"> <img src="img/party3.png"> Party C </button> <button name="party" value="Party D"> <img src="img/party4.png"> Party D </button> </form> <a href="results.php">View Results</a> </body> </html>

results.php	<?php
	include "db.php";
	?>
	
	<!DOCTYPE html>
	<html>
	<head>
	<title>Voting Results</title>
	</head>
	
	<body>
	
	<h1>Voting Results</h1>
	
	<?php
	
	$parties = ["Party A","Party B","Party C","Party D"];
	
	foreach($parties as $p){
	
	$result = $conn->query("SELECT COUNT(*) as total FROM votes WHERE party='$p'");
	$row = $result->fetch_assoc();
	
	echo "<h3>$p : ".$row['total']." votes</h3>";
	
	}
	
	?>
	
	<a href="index.html">Back to Vote</a>
	
	</body>
	</html>

style.css	body{
	font-family: Arial;
	text-align:center;
	background:#f4f4f4;
	}
	
	.container{
	margin-top:50px;
	}
	
	button{
	width:150px;
	height:150px;
	margin:10px;
	font-size:18px;
	cursor:pointer;
	}
	
	img{
	width:60px;
	}

vote.php	<?php include "db.php"; if(isset($_POST['party'])){ $party = $_POST['party'];

	$sql = "INSERT INTO votes (party) VALUES ('$party')"; $conn->query($sql); echo "<h2>Thank you for voting!</h2>"; echo "<a href='results.php'>See Results</a>"; } ?>
Image files here	img/party1.png img/party2.png img/party3.png img/party4.png